Magnetide — Build Instructions v21-r2i

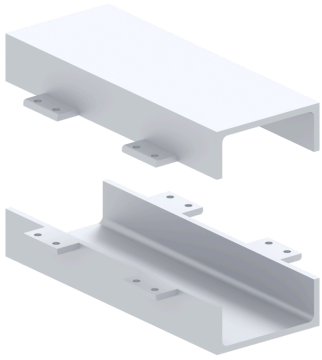
MHD seawater pump · 150 × 81 × 45 mm
Kelas 10.8, Kelompok 1 · Reynard Filbert Tang
Rev r2i · 18 parts · 7 steps

2×	2×	2×	96×	2×	1+1	8×
YOKE	TRAY	CAP	MAGNET	ELECTRODE	ENDS	M3 BOLT
3.0 mm steel · 150 long	124 × 50.7 × 19.5	1 mm · Ø3 pegs	20 × 10 × 5 N52	graphite 134 × 14 × 2	bellmouth · rear block	M3 × 14 + nut

1 Fold the two yokes

Bend 90° **down** on the **R3** line (dashed, at ±25.89) for the walls, then the two lugs **out** on the **R1.5** line at ±43.50. One blank per half, mirror image.

> Check after bending: 57 mm across the plate, 22.5 mm plate face to lug face, lug 12 mm, hole 8.25 mm off the plate face, in the flat past the bend. Two Ø3.4 holes per lug, 13 apart, at x 40 / 110. R3 is not optional on 3 mm sheet — a sharp die splits the outside of the bend.

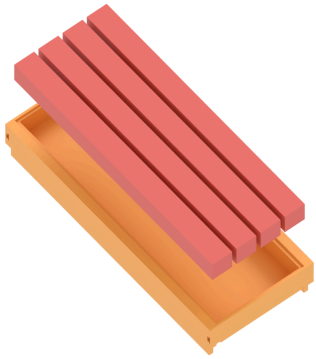


Need: 2 × clamshell-flat-3.0mm.dxf, folded · **Then:** deburr all edges

2 Load the magnets

48 tiles per tray: 6 along the bore, 4 across the width, 2 layers deep. They drop into the cavity and finish flush with the floor.

> **Compass check before the epoxy sets.** All four rows on a face point the same way; the other tray points the other way. One flipped row stalls a quarter of the bore.

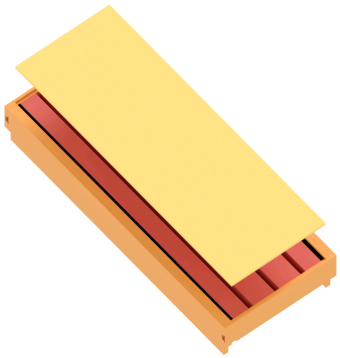


Need: 2 × tray, 96 × N52 tile, epoxy · **Drawn as 4 bars** = the 48 tiles of one tray · drop in, never press-fit

3 Cap it

Drop the 1 mm cap on the ledge. It sits on the magnets and finishes flush with the steel inner face.

> This face is the magnet's road into the yoke. A bead of epoxy at the rim only — **never** a puddle.

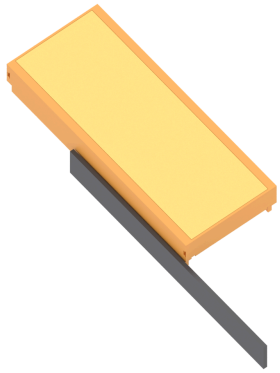


Need: 2 × cap · **Result:** stack height 19.5 mm per half

4 Slide in the electrodes

Each graphite plate slides into the 1 mm groove from the end. Push until it sits 5 mm inside the end pocket.

> Full 14 mm face exposed: 124 × 14 = 1736 mm² per plate. Keep the edges clean — a burr on graphite is a short waiting to happen.

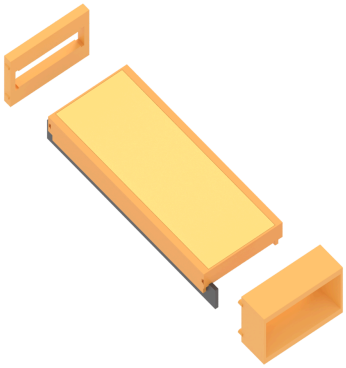


Need: 2 × electrode · **Fit:** 0.2 mm clearance in the groove

5 Bellmouth and rear block

Two Ø3 pegs each, into the blind holes in the tray end faces. Mouth in, straight end out.

> **Wire it now:** drill through the last 0.4 mm of the lead channel, feed the two wires out, epoxy the hole shut around them.

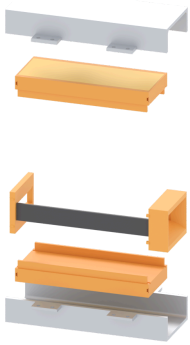


Need: 1 × bellmouth, 1 × rear block, epoxy · **Keep:** the outlet straight

6 Close the yoke

Lower tray into the bottom yoke, upper into the top. Bring the halves together so the lug faces meet bare — no paint, no gasket.

> This is a **stack clamp**: the tray rim lands on the steel plate, the mirrored tray butts it at mid-plane, the bolts close the sandwich.



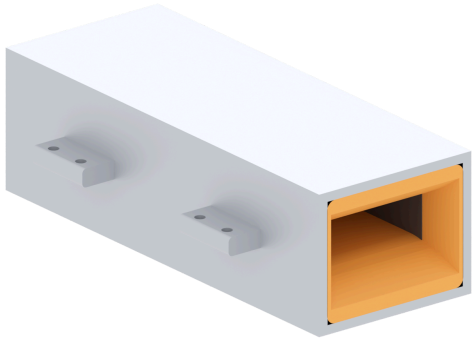
Need: 2 × assembled half · **Check:** split faces bare metal

7 Bolt it

Eight M3 × 14 through the lugs — **two per lug, 13 mm apart**. Tighten in a cross pattern, a little at a time, so the halves seat flat instead of rocking.

> 2 N·m gives roughly 8 kN across the four lug pairs. Run a nut and washer on the underside; thread never protrudes into the bore.

Need: 8 × M3 bolt, nut, washer · **Check:** no gap at any lug face



Before you power it

- ☐ Both halves flush, 8 bolts seated, no gap at the lug faces
- ☐ Compass: every row on a face points the same way, the two faces oppose
- ☐ Tray floor unbroken over the whole magnet area — hold it up to a light
- ☐ Electrodes fully home, not shorting to steel through a scratch
- ☐ Lead exit drilled, wired, epoxied; bore clear down its whole 124 mm
- ☐ Fuse in the supply line; runs under 2 minutes, ventilated (Cl_2)

If it does not fit	Nominal	Cause
tray rattles sideways	0.15 / side	shim 0.3–0.5 mm closed-cell foam on the **outer** wall only
cap stands proud of the steel	1.0	epoxy puddle under the cap, or a magnet sitting proud
electrode will not enter	2.0 in 2.2	graphite burr, or the groove has epoxy in it
halves will not close	0 gap	tray rim not seated on the plate — check the stack, not the bolts